

CODE SPORT



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In this month's column, we discuss the topic of natural language understanding in the context of dialogue systems.

In last month's column, we had discussed the use of universal sentence embedding representations. We also briefly talked about using these representations to cluster a given set of documents. In this month's column, we discuss a closely related topic — how we can apply natural language understanding to a corpus of documents to understand the nature of topics and queries in that corpus. We discuss natural language understanding in the context of dialogue systems.

Conversational voice agents are becoming extremely popular. Apple Siri, Google Assistant and Amazon Alexa have been widely adopted. If you wondered about the natural language capabilities needed to build a conversational agent, we plan to cover this topic in this column over the next couple of months.

There are two basic types of dialogue or conversational agents. These are the open domain dialogue agents versus the closed domain dialogue agents. Closed domain dialogue agents or chatbots are typically restricted to conversations in one specific domain. Consider a chatbot that is designed to help you with your banking transactions. This is a domain-specific chatbot. Closed domain chatbots are typically focused on helping the user in a specific task. Hence they are typically task-oriented or goal-oriented.

On the other hand, there are open domain chatbots with which you can have conversations on virtually any topic. Apple Siri is an example. Open domain chatbots are difficult to design since they need to handle any kind of conversation. Task-oriented or goal-oriented chatbots typically have their conversations focused on one domain — for example, hotel reservations or flight ticket booking. On the other hand, chit-chat dialogue systems are those with which users can engage in general conversations such as “How are you feeling today?” or “What is your favourite colour?”. Since chit-chat conversations make chatbots seem more human, today, most task-oriented chatbots also have some capability to handle chit-chat conversations.

You may be wondering in what way natural language understanding is a critical capability in the design of chatbots. Let us look at an example of a task-oriented chatbot, which

allows you to make hotel or restaurant reservations. Now the user interacts with the chatbot with the utterance: “Get me a table in the best Chinese restaurant in East Bangalore.” This specific utterance needs to be mapped to the action or task of booking a table in a restaurant. This is known as ‘intent’. An intent captures the specific action/task expressed in a user utterance. Intents are typically pre-defined by the dialogue system designer and map to the various actions that are typically requested by the user. If the user says “Please cancel my booking in Empire Restaurant for tonight,” then the intent for that utterance is *Cancel_booking*.

Now what happens if the user comes up with an utterance that has no corresponding intent defined by the dialogue designer? In order to avoid this problem, typically, a default intent is defined by the dialogue designer, which will catch all the unknown utterances from the user. For example, if the user says, “Book me a movie ticket” to the above restaurant-booking chatbot, there may not be any intent defined for booking movie tickets. Hence this will get mapped to the default intent and the dialogue system will prompt the user for further information by saying, “Sorry, I don’t understand your request. Can you please state your request again or do you want to talk to a human agent?”

Now you may be wondering how a user utterance gets mapped to a specific intent. This is the job of the natural language understanding (NLU) module of the dialogue system. Given user utterances, the NLU module identifies which of the predefined intents the particular utterance can be mapped to. It also tries to identify the entities defined in the utterance. Given the example utterance, “Get me a table in the best Chinese restaurant in East Bangalore,” there are three slots/entity types — rating, cuisine type and location which have the corresponding values: ‘best’, ‘Chinese’ and ‘East Bangalore’. Hence, given a user utterance, the NLU module needs to identify not only the entity types and their values but also the intent. Let us first look at the problem of intent identification.

Intent identification can be considered equivalent to the problem of paraphrase identification. For each predefined intent, users have multiple ways of expressing that intent in a